

Notes that by invariance of dim. is a shorthand for $|I| \leq |S|$ where I is lin. independent and S a spanning set.

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Problem 1

(a)

(1) claim: the vectors $(1 \ 2 \ 0 \ -1)^T$ and $(0 \ 0 \ 1 \ 3)^T$ are form a-basis of the row space of A .

Proof - Note that elementary row operations preserve the row space of A . In particular since elementary row operations only scale, swap and make linear combinations of the row vector, the span is unaffected:

- Swapping obviously preserves span
- Scaling preserves span because for any $\underline{x} = a_1 \underline{r}_1 + a_2 \underline{r}_2 + a_3 \underline{r}_3$,

$$\underline{x} = \frac{a_1}{c} (c \underline{r}_1) + a_2 \underline{r}_2 + a_3 \underline{r}_3$$

where $\underline{r}_i$ are the vectors, $\underline{x} \in \text{span}(\underline{r}_1, \underline{r}_2, \underline{r}_3)$ and $c \neq 0$ by definition.

- Adding $a_i \underline{r}_i$ to $\underline{r}_j$ does something similar: (the following is w.l.o.g.)

$$\underline{x} = a_1 \underline{r}_1 + a_2 \underline{r}_2 + a_3 \underline{r}_3$$

$$\Rightarrow \underline{x} = a_1 (\underline{r}_1 + a_2 \underline{r}_2) + (a_2 - a_1 a_2) \underline{r}_2 + a_3 \underline{r}_3$$

$$\text{Thus, } \text{span}((1 \ 2 \ 0 \ -1)^T, (0 \ 0 \ 1 \ 3)^T, (0 \ 0 \ 0 \ 0)^T)$$

$$= \text{span}(\underline{r}_1, \underline{r}_2, \underline{r}_3) = \text{Row space of } A.$$

Now note that $\text{span}((1 \ 2 \ 0 \ -1)^T, (0 \ 0 \ 1 \ 3)^T)$ = Row space of A as the zero vector contributes nothing.

Thus, And $(1 \ 2 \ 0 \ -1)^T, (0 \ 0 \ 1 \ 3)^T$ are linearly independent:

$$c_1 \begin{pmatrix} 1 \\ 2 \\ 0 \\ -1 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} c_1 \\ 2c_1 \\ c_2 \\ 3c_2 - c_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow c_1 = c_2 = 0.$$

By defn, they form a basis.

- (2) For a general matrix A , we convert it to its RREF. Then, the non-zero rows (or the pivot rows) form a basis of the row space.

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(3) Note first that again, the row space of A is unaffected by the elementary row operations (as we've argued in class, and I argued in a less general sense in (1)). Thus, and because the zero rows add nothing, we have that the non-zero rows span the row space of A . We claim that these are also linearly independent.

Let these be $\underline{r}_1, \underline{r}_2, \dots, \underline{r}_n$. Consider:

$$c_1 \underline{r}_1 + c_2 \underline{r}_2 + \dots + c_n \underline{r}_n = \underline{0}$$

Consider ~~there~~ let the pivot column of $\underline{r}_i$ be in the K_i column (or co-vector, in K_i ~~co-ordinates~~) then this entry will be zero everywhere else, ~~and that~~ (if zero for all other $\underline{r}_j$ for $j \neq i$) by definition of RREF.

Then, the K_i entry with ~~of~~ the linear combination ~~co-ordinates~~ will result in the equality $c_i = 0$.

Hence $c_i = 0 \quad \forall i \Rightarrow \{\underline{r}_i\}$ are linearly independent.

By definition of a basis, we are done.

(b)

(1) The vectors $\begin{pmatrix} a_{11} \\ a_{12} \\ a_{13} \end{pmatrix}$ and $\begin{pmatrix} a_{31} \\ a_{32} \\ a_{33} \end{pmatrix}$, ~~form~~ the pivot columns

of ~~A~~ A 's RREF form the basis of $\text{Im}(f)$. If

(2) convert A to its RREF. ^{using row operations.} Then, read off the columns which have pivots. If the $k_1, \dots, k_m$ columns of A_{RREF} contain pivots, then $\{v_{k_1}, \dots, v_{k_m}\}$ form a basis of the image of f .

(3) First note that

$$f(x) = Ax = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \dots + x_m \begin{pmatrix} a_{1m} \\ \vdots \\ a_{mm} \end{pmatrix}$$

by vector-matrix multiplication. Thus, ~~A~~ the column vectors of A span the image of f [since $v \in \text{Im}(f)$]

$$\Rightarrow \exists x = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} : f(x) = v = x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \dots + x_m \begin{pmatrix} a_{1m} \\ \vdots \\ a_{mm} \end{pmatrix}$$

Now we claim that row operations don't change the dependency relations b/w columns. i.e. if column vectors $k_1, \dots, k_p$ are independent before a row operation, then they're independent after one.

- swap just shifts the co-ordinates, no nothing changes.
- scaling ^{in rows} scales each k_i (column's ⁱⁿ co-ordinates by the same amount, so again nothing changes
- $[c_1 a_1, c_2 a_2, \dots, c_p a_p] \neq 0$ now would imply the c_i are not all zero. ~~non-zero~~ times a linear combination of the i^{th} coordinates is 0 $\Rightarrow$ not independent

~~What does A do to E_{ij} ?~~

~~$$\begin{aligned}
 A L(E_{ij}) &= A E_{ij} \\
 &= \begin{pmatrix} a_{11} & \dots & a_{1i} \\ & \ddots & \\ & & a_{nn} \end{pmatrix} \begin{pmatrix} 0 & \dots & 1 & \dots & 0 \\ & \ddots & & \ddots & \\ & & 0 & \dots & 0 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & \dots & a_{1i} & \dots & 0 \\ & \ddots & & \ddots & \\ & & 0 & \dots & 0 \\ & & & \ddots & \\ & & & & 0 \end{pmatrix}
 \end{aligned}$$

(1)st column~~

or in terms of co-ordinates:

~~$$\begin{pmatrix} 0 \\ a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \\ 0 \end{pmatrix} \leftarrow \begin{matrix} j \\ j+n \end{matrix} \leftarrow j \text{th column}$$~~

~~Thus the matrix of L looks like~~

~~Thus, based on the recipe we did in class for finding the matrix for two abstract vector spaces,~~

~~$$[L(x)]_{E_{ij}} = L \cdot [x]_{E_{ij}}$$~~

~~where L is an $n \times n$ matrix, constructed as the i -th column~~

~~of L being $L E_{ij}$.~~

- Finally, adding a multiple of the j th row to the i th row also doesn't change ~~it~~ it as we covered in the tutorial.

$$\begin{aligned}
 & \left(\begin{array}{c} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{array} \right) + b_j \left(\begin{array}{c} a_{j1} \\ a_{j2} \\ \vdots \\ a_{jn} \end{array} \right) \\
 &= b_1 \left(\begin{array}{c} a_{11} \\ a_{12} \\ \vdots \\ a_{1n} \end{array} \right) + \dots + b_p \left(\begin{array}{c} a_{p1} \\ a_{p2} \\ \vdots \\ a_{pn} \end{array} \right) + \left(\begin{array}{c} a_{i1} \\ a_{i2} \\ \vdots \\ a_{in} \end{array} \right)
 \end{aligned}$$

Now note that the pivot columns of the RREF must be linearly independent, since pivot columns of the RREF have 1 in the k_i co-ordinate and the rest are 0, so the linear combination can only be zero if the weights are 0. This implies the pivot columns of A are lin. independent.

Note that since independence is preserved across row operations, as they're invertible, we argue that ~~the~~ ~~column~~ ~~except~~ the pivot columns are maximally linear independent, i.e. $\exists x \in \text{Im}(A)$ s.t. $\{\text{pivot columns}\} \cup \{x\}$ is also linearly independent.

Clearly if this is lin. independent, and column vectors span $\text{Im}(A)$, $\exists$ a column vector $\in \phi \setminus \{\text{pivot columns}\}$ s.t. $\{\text{pivot columns}\} \cup \{x\}$ is lin. independent.

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But if that were true, it'd be true in the RREF too. However if the column is not a pivot column in the RREF, if either is zero, or shares ~~non-zero~~ non-zero co-ordinates with some pivot column. Thus by choosing weights we can

form a lin. combination of $\{ \text{pivot column} \cup \{e_i\}$ that is $\underline{0}$, which violates lin. independence. Since we showed in class that a maximal lin. independent set forms a basis, we are done.

(c)

(1) The basis vectors:

$$\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -3 \end{pmatrix}$$

form a basis of the kernel.

(2)

convert A to its RREF. Then mark the columns that do not have any pivots. Let these be $e_1, \dots, e_{29}$. ~~Then~~ Then the vectors

$$\text{The vector } x_1, \dots, x_{29}$$

form a basis where the i th co-ordinate of x_m is 1 if $i=m$, the ~~the~~ negative of the ~~i th~~ ^{m th} entry in the ~~i -th~~ ^{i -th} row, if i th row has a pivot and zero otherwise.

(3) Note that if

$$\underline{a} = c_1 x_1 + \dots + c_m x_m$$

by the above recipe we read off the i th entry in the i th row for the i th coordinate forming a linear combination of them ~~gives~~ ~~the~~ ~~precisely~~ the equation ~~$c_1 a_{i1} + \dots + c_m a_{im} = 0$~~ ~~precisely~~ the equation $A \underline{a} = \underline{0}$ gives $a_i \neq 0$ if i is not a pivot column

2.

Consider the basis:

$$\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ of } \mathbb{R}^4.$$

It is easy to verify that this is indeed a basis.

For the basis of $\mathbb{R}^3$, consider the basis formed by ~~or change~~ $v_1 = f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right)$, $v_2 = f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right)$, ~~eg~~ and extending by

another vector e_1, e_2, e_3 by checking which is independent to both of them. Claim: this will form a basis.

Note that $f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right)$ and $f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right)$ are independent as:

$$c_1 f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right) + c_2 f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \cancel{f\left(\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right)} \neq \cancel{f\left(\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right)}$$

$$\Rightarrow f\left(\begin{pmatrix} 0 \\ c_2 \\ c_1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

from our work in P1, we can see that

$$f\left(\begin{pmatrix} 0 \\ c_2 \\ c_1 \end{pmatrix}\right) = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow R \cdot \begin{pmatrix} 0 \\ c_2 \\ c_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \text{ which clearly cannot be.}$$

for non-zero c_1, c_2 . Now one of c_i must be independent to both, otherwise $\text{span}(v_1, v_2) = \text{span}(e_1, e_2, e_3)$ violates invariance of dimension, which we showed in class.

Again by invariance of dimension, since v_1, v_2, e_i are lin. independent, they must be maximally lin. independent and hence span $\mathbb{R}^3$.

or a_{ij} is a pivot, i.e. 1.) Hence if the i th row has a pivot, we get the equation obtained from the i th row of $A_{REF} \underline{x} = 0$. If it doesn't, then we either get $a_{ii} = x_{ii}$ or $x_{ii} = 0$ for the non-pivot rows, as should be the case.

Thus all linear combinations of these vectors are in $\ker(f)$. Note that they're lin. independent because the non-pivot i th entry (where it does not have a pivot) is non-zero iff only for one $\underline{x}_i$, so these are the linear combination being 0 would imply that all weights are zero

Anyhow, note that all $\underline{x} \in \ker(f)$ are representable in this manner, because if they're not, then they will not follow the equations read off from $A_{REF} \underline{x} = 0$. Finally note that $f(\underline{x}) = 0 \Leftrightarrow A_{REF} \underline{x} = 0$ as we've shown in class,

so these vectors do indeed form a basis of $\ker(f)$.

Note that R is indeed the REF of A (by uniqueness, as shown in tutorial, and inspection).

By the recipe we did in class, ~~this~~ the way to construct A' is that the i th column must be ~~the~~ $f(w_i)$ where w_i form an ordered basis of $\mathbb{R}^4$.
 Ordering our basis as $w_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, $w_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $w_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 0 \end{pmatrix}$, $w_4 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 3 \end{pmatrix}$
 and the basis in $\mathbb{R}^3$ as

$$u_1 = f \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, u_2 = f \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, u_3 = e_i$$

Then,

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (\text{especially as } f(w_3) = f(w_4) = 0).$$

which has no non-zero diagonal entries, so we are done.

(b) We claim that this is the number of pivot ~~rows~~ rows.

Suppose A is a matrix representing f . Let v be

$$[f(v)]_C = A[v]_B.$$

Then a change of basis, $C \mapsto C'$, $B \mapsto B'$ is done by the change of basis formula we proved in class:

$$A' = Q^{-1} A P \quad \text{where } [f(v)]_{C'} = A'[v]_{B'}.$$

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Now, note that Q and P are invertible as we proved in class. However, these ~~are~~ compositions of elementary matrices, as we also proved in class.

But since multiplication by elementary matrices on the left is equivalent to elementary row operations, and on the right, to elementary column operations, any representation of f is achieved through these operations. As we showed in the tutorial any matrix A can be transformed to a top-right identity matrix through these operations.

$$\begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 1 \end{pmatrix} = I_m$$

3 where r is the rank or the number of pivot rows of the matrix. Note that if some other representation has fewer non-zero terms, by the change of basis to that representation would involve elementary and column operations, these operations must kill of a term in this matrix which is not possible! hence we are done.

Answer 3

(b) if $f \circ g = \text{id}_W$, then note that f must be ~~an~~ ~~is~~ a surjection, because $f \circ g = \text{id}_W \Rightarrow \forall x, f \circ g(x) = x$

$$\Rightarrow \forall x, \exists y, f[g(x)] = x$$

$$\Rightarrow \exists y, \forall x, \exists y: f(y) = x$$

~~But, we want to show that f is a representation of $\mathbb{R}$~~

~~Let R~~ By the rank-nullity theorem:

$$\dim(\ker(f)) + \dim(\text{Im}(f)) = \dim(V)$$

$$\Rightarrow \dim(\ker(f)) + \dim(W) = \dim(V)$$

Hence, $\dim(V) \geq \dim(W)$ with equality iff $\dim(\ker(f)) = 0$

$$(\Rightarrow) \ker f = \{0\}.$$

But if $\ker f = \{0\} \Rightarrow f$ is injective as we have shown in class. Since f is also surjective, it must be bijective!

~~But then, f must be a left inverse. ~~So~~ too. ~~Let this be~~~~
~~shown. ~~But then,~~~~

$$\cancel{R = \text{is id}}$$

But then, f has a unique inverse! Therefore $|R| = 1$ violating our assumption. Thus, $\dim(V) > \dim(W)$.

(a) Consider the equivalence classes of V ,
 $[V] = \{ \text{~~the~~ } u \in V : f(u) = f(v) \}.$

Clearly this is an equivalence relation.

Note ~~the~~ now that for any g ,

$$g(x) \in [g_0(x)], \text{ because } f(g(x)) = f(g_0(x)) = x$$

I am not sure how to better do this.

In particular consider $\text{Im}(g)$, then for any g ,

$\text{Im}(g)$ must be the same under this equivalence relation, i.e. $[g(x)] = [g_0(x)]$ for all $x \in W$.

Problem - 4

(a) Let A be the matrix representing the linear map T with respect to basis C and B be the representation of T wrt basis C' . Let the change of basis matrix be P such that $C' = PC$. As we have shown in class,

$$B = P^{-1}AP.$$

Now, $\det(B) = \det(P^{-1}AP).$

Recall as we showed that $\det$ is multiplicative,

$$\begin{aligned} \det(B) &= \det((P^{-1}A)P) = \det(P^{-1}A)\det(P) \\ &= \det(P^{-1})\det(A)\det(P) \end{aligned}$$

Since $\det(M) \in \mathbb{F}$, a field for all M , this ~~is~~ it commutes wrt multiplication so that:

$$\begin{aligned} \det(B) &= \det(A)\det(P^{-1})\det(P) \\ &= \det(A)\det(P^{-1}P) \quad \leftarrow \text{by multiplicativity of } \det. \\ &= \det(A)\det(I) \\ &= \det(A) \cdot 1 \\ &= \det(A) \end{aligned}$$

Since $\det(I) = 1$ by definition.

Thus

$$\det(B) = \det(A)$$

regardless of the choice of basis. So we are done.

(6) ~~other~~

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(b) Obviously this is a little ~~handy~~ but we first note that $M_{n \times n} \subseteq \mathbb{R}^{n^2}$ by the map $M_{ij} \mapsto$ a matrix M maps to a vector m : $M_{ij} \mapsto m_{(i-1) \cdot n + (j-1)}$

(lexicographically). The point is that we have a basis now ~~however~~ we can write down the matrix, wrt basis: $\{E_{ij}\}$, the matrix units. Note that there are linearly independent as:

$$\text{if } M = \sum_{i,j} c_{ij} E_{ij}$$

Then $M_{ij} = c_{ij}$ (as all the other matrix units have zero here)

Thus, $\sum c_{ij} E_{ij} = 0 \Rightarrow c_{ij} = 0 \forall i,j$. So they're linearly independent. But note that they also span $M_{n \times n}$ as:

$$M = \sum M_{ij} E_{ij}$$

Thus they form a basis (with ~~lexicographic~~ lexicographic ordering).

Let us now represent L with this basis

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Then based on our work

$$d = \begin{pmatrix} a_{11} & 0 & \dots & 0 \\ 0 & a_{11} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & a_{nn} \end{pmatrix}$$

$\{n\}$

$u(b)$.

Consider the basis formed by extending the basis of $\ker(L)$. Then note that ~~by choosing~~ if $\dim(\ker(L)) > 0$, ~~ie there is a row where~~ then $\exists v_i : d(v_i) = 0$ in this basis. Hence, by our recipe of forming the matrix of d , we have that one (or more) rows will be entirely 0. Note that then,

$$\det(L) = \det(c_1, \dots, c_n) \quad \text{where } c_i \text{ are}$$

columns of representing of L w.r.t. this basis

$$= \det(0, \dots, c_n) \quad \text{wlog}$$

$$= \det(0+0, \dots, c_n)$$

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$$= \det(0, \dots, c_n) + \det(0, \dots, c_n)$$

$$= 2 \det(L)$$

$$\Rightarrow \det(L) = 0. \quad \text{Hence } \det(L) \neq 0 \Rightarrow$$

$$\ker(L) = \{0\}. \quad \text{Also note that if the kernel is}$$

See exam choice

$$\Rightarrow AX=0 \text{ if } X=0_{n \times n} \text{ matrix}$$

but if ~~$\det(A) \neq 0$~~ , then ~~$\det(AI_n) = \det(A) \neq 0$~~ note if we convert to $\det(AI_n)$ then $\det(A) \neq 0$

Then there we see

consider $\text{REF}(A)$. we claim it has a free non-pivot column.

$$\text{since } A_{\text{REF}} = E_1 \cdots E_k A,$$

$$\det(A_{\text{REF}}) = \det(E_1) \cdots \det(E_k) \det(A) = 0$$

$\Rightarrow A_{\text{REF}}$ cannot be I , so it must have non-pivot rows and thus columns (because it is square).

But then, since there is a free column

$$\Rightarrow \exists \text{ non trivial } X: AX=0$$

$$\text{violating that } \ker(A) = \{0\}.$$

Hence

$$\det(L) \neq 0 \Rightarrow \det(A) \neq 0.$$

For the other way around, note that $\det(E_i) \neq 0$ for elementary matrices, because scaling $\Rightarrow \det(E_i) = c \cdot \det(I)$. swap $\Rightarrow \det(E_i) = -\det(I)$. Adding rows $\Rightarrow \det(E_i) = \det(I)$.

$$\text{Thus } \det(A_{\text{REF}}) \neq 0 \Leftrightarrow \det(A) \neq 0.$$

In particular, since A is square, and $A_{\text{REF}} = I$ (otherwise, $\det$ will be zero), $\Rightarrow A_{\text{REF}}$ has an inverse as proved in class.

Therefore

$$\det(A) \neq 0 \Rightarrow A \text{ has an inverse.}$$

But then, ~~Assume~~ $\ker(A)$ must be 0, as

$$\begin{aligned} A[x]_{\{e_{ij} \text{ basis}\}} &= [0]_{\text{basis}} \\ \Rightarrow [x]_{\{e_{ij}\}} &= A^{-1}[0]_{\{e_{ij}\}} \\ &= [0] \end{aligned}$$

$$\text{Hence } \ker(A) = \{0\}.$$

Now note that since A has an inverse, we can find

$$v_1, \dots, v_n:$$

$$A[v_i] = [e_{ii}]$$

$$\text{by simply using } [v_i] = A^{-1}[e_{ii}]$$

But now note that these v_i 's must be lin. independent?

$$c_1 v_1 + \dots + c_n v_n = 0$$

$$\Rightarrow f(c_1 v_1 + \dots + c_n v_n) = 0$$

$$\Rightarrow c_1 [e_{ii}] + \dots + c_n [e_{nn}] = 0 \text{ which cannot be.}$$

By invariance of dimension, ~~they must~~ they form a basis.

$\hookrightarrow$ they must be a maximally lin. independent set, by (lin independent $\leq$ span)

But then, representing A in this basis ~~gives us identity~~ ~~matrix, so that the det would be 1.~~

~~$$\det(A) \neq 0 \Rightarrow \det(I) \neq 0.$$~~

Note that e_{ij} ,
the matrix units,
form a basis of
 M_{nn}

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Now consider the matrix representation of L w.r.t $\{e_i\}_i = \mathcal{M}$

Suppose $\det(\mathcal{M}) = 0$. By our prev. argument,

$\det(\mathcal{M}_{\text{REF}}) = 0 \Rightarrow$ the columns are not all independent. But each column is $A(\tilde{e}_{ij})$, based on our recipe of constructing matrices, so them not being linearly independent would imply

$$\sum c_{ij} A \tilde{e}_{ij} = 0$$

$$\Rightarrow A(\sum c_{ij} \tilde{e}_{ij}) = 0$$

Note that since $\ker(f) = \{0\}$

$\Rightarrow \sum c_{ij} \tilde{e}_{ij} = 0$ but since $\tilde{e}_{ij}$ form a basis

$\Rightarrow c_{ij} = 0 \forall i, j$, therefore the REF cannot have zero det!

Hence $\det(\mathcal{M}) \neq 0$.

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(5)

(b)

Let $M = (c_1 \ c_2 \ \dots \ c_{2026})$ where c_i are column vectors.

Then $-M = (-c_1 \ \dots \ -c_{2026})$.

Since $\det$ is multilinear:

$$\begin{aligned} \det(-M) &= \det(-c_1 \ \dots \ -c_{2026}) \\ &= -1 \cdot \det(c_1 \ -c_2 \ \dots \ -c_{2026}) \\ &= (-1)^2 \cdot \det(c_1 \ c_2 \ -c_3 \ \dots \ -c_{2026}) \\ &\vdots \\ &= (-1)^{2026} \det(c_1 \ c_2 \ \dots \ c_{2026}) \\ &= 1 \cdot \det(M) \\ &= \det(M) \end{aligned}$$

• $2M = (2c_1 \ 2c_2 \ \dots \ 2c_{2026})$

Then again using multilinearity:

$$\begin{aligned} \det(2M) &= \det(2c_1 \ 2c_2 \ \dots \ 2c_{2026}) \\ &= 2 \cdot \det(c_1 \ 2c_2 \ \dots \ 2c_{2026}) \\ &= 2^2 \det(c_1 \ c_2 \ 2c_3 \ \dots \ 2c_{2026}) \\ &\vdots \\ &= 2^{2026} \det(c_1 \ \dots \ c_{2026}) \\ &= 2^{2026} \det M \end{aligned}$$

• M st. $m_{ij} = 0$ if $i < j$.

By the determinant formula we showed in class:

$$\det M = \sum_{\sigma \text{ is a bijection}} \operatorname{sgn}(\sigma) M_{1\sigma(1)} \cdots M_{n\sigma(n)}$$

Note that since $M_{1j} = 0$ if $1 < j$, the sum is zero for all $\sigma : \sigma(1) \neq 1$. Thus the only part that survives is:

$$\sum_{\sigma \text{ is a bijection}} \operatorname{sgn}(\sigma) M_{11} M_{2\sigma(2)} \cdots M_{n\sigma(n)} \quad \sigma(1) = 1$$

where now $\sigma(i) \neq 1 \forall i \neq 1$. Similarly $\sigma(2)$ can only be 2 or 1. Since it cannot be 1, we have that $\sigma(2) = 2$. for the non-zero terms.

Therefore, inductively we must have that:

$$\det(M) = \sum_{\sigma \text{ is a bijection}} \operatorname{sgn}(\sigma) M_{11} M_{22} \cdots M_{nn}$$

$$\sigma(1) = 1,$$

$$\sigma(2) = 2,$$

$$\vdots$$

$$\sigma(n) = n$$

$$\det(M) = m_{11} m_{22} \cdots m_{nn}.$$

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$\operatorname{sgn}(\sigma) = 1$ as there is no ~~extra~~ inversion done.

(c) Claim: ~~The dimension of~~ V ~~must be~~ $\geq \dim L$ ~~or~~ the list L must be lin. independent.

We prove this by showing that the v_i 's must be linearly independent. Suppose not, then $\exists c_1, \dots, c_n$, not all zero such that

$$f(c_1 v_1 + \dots + c_n v_n) = 0.$$

But,

$$f(c_1 v_1 + \dots + c_n v_n) = c_1 f(v_1) + \dots + c_n f(v_n)$$

$$f(0) = c_1 x_0 + c_2 x^2 + \dots + c_n x^n$$

$$0 = c_1 x + c_2 x^2 + \dots + c_n x^n$$

$$\left. \begin{array}{l} \text{For a linear map} \\ f(0) = 0 \text{ as } f(x+0) = \\ f(x) + f(0) = f(x) \Rightarrow f(0) = 0 \end{array} \right\}$$

Since atleast one of c_i 's are non-zero, we have that a deg ≥ 1 polynomial is equal to 0 which cannot be!

Note that this is also sufficient, if L is independent, the map $f(v_i) = x^i$ is linear as ~~is~~ ~~can be made linear~~ is linear, as we may note that we can do a map like:

$$\begin{array}{ccc} \mathbb{R}^m & \xrightarrow{\quad} & \mathbb{R}^n \\ \uparrow \text{is} & & \downarrow \\ \mathbb{V} & & \mathbb{R}[x] \end{array}$$

(coefficient of
max n -degree polynomial)

~~Q2~~ (a) It shows us that the dimension of the row and column spaces are the same.

One can do it as follows. The rows span $\mathbb{R}^q \Leftrightarrow$ row space is ~~$\mathbb{R}^q$~~ or $\mathbb{R}^m$ where $m \geq q$ $\Leftrightarrow$ column space is $\mathbb{R}^m$ where $m \geq q$, but this since we have q columns, at max. we can have that the column space is $\mathbb{R}^q$, when all columns are independent.